

Dear Hiring Manager,

I am writing to express my interest in the Senior Geospatial Machine Learning Scientist position at Outbreak Labs. Having built spatial ML platforms from 0 to 1 - most recently architecting nationwide diffusion-based forecasting models at Homecastr and engineering a statewide high-resolution elevation platform at Summit Geospatial - I am now seeking to bring that execution speed to a deeply technical, mission-driven team. Your goal to mitigate agricultural pests perfectly aligns with my background in Earth observation.

Prior to my current ventures, I spent five years as a Senior Data Scientist at the Texas Advanced Computing Center (TACC). There, I gained firsthand experience with the complexities of agricultural and environmental modeling by competing in the DARPA World Modelers program to forecast food security in East Africa - work that led to joint publications on intelligent systems with the former president of AAIL. I also led the data pipelines for the \$40M Texas Disaster Information System, scaling models across leadership-class supercomputers and contributing fundamental geospatial research that supported the 2022 EGU Bagnold Medal in geomorphology.

My technical stack spans both modern generative AI workflows and the core geospatial libraries you require, including Python, PyTorch, TensorFlow, GeoPandas, Rasterio, GDAL, and xarray. As an early-stage founder, I thrive in fast-paced environments where I can take ownership of both the research direction and production engineering. I am excited by the opportunity to apply my expertise to build data-driven strategies that protect global agricultural production.

I would welcome the opportunity to discuss how my background can contribute to Outbreak Labs' mission. Please note that regarding work authorization, I am eligible to independently secure the UK Global Talent Visa (Tech Nation) without employer sponsorship. To accommodate a fast-moving start date, I am also open to beginning the role as a US-based independent contractor while my visa processes for relocation.

Sincerely,

Daniel Hardesty Lewis

Attached: curriculum vitae

Daniel Hardesty Lewis

Geospatial ML Scientist – earth observation, distributed infrastructure, computer vision

Specializing in automated validation frameworks and scalable distributed infrastructure. Delivered consumer pricing and statewide resiliency platforms across geospatial & time-series workloads.

Core Expertise

- 8 years **ML Engineering**: Build and operate AI/ML training and inference pipelines, using rigorous validation protocols to ensure model generalization and reliability
- 8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS
- 8 years **Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs
- 10 years **Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases
- 8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

Technologies

Modeling	PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT
LLM apps	embeddings; RAG; LangChain; LangGraph; DSPy; FAISS
Data stack	Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase
Infrastructure	Linux/Unix; Bash; Git; Docker; Kubernetes; Modal; GCS; Ray; Spark; Dask; Airflow; CI/CD; observability
Additional	CUDA; C++; TypeScript; React

Experience Professional

- Nov. 2024 – Present **Co-founder**, [Homecastr](#), New York, NY
Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation.
 - Architected a nationwide tract-level diffusion forecasting model, **beating Zillow's 8.4% one-year error benchmark at 8%** and maintaining a **stable 25% MdAE over a 4-year horizon**.
 - Engineered architecture incorporating **spatial inducing tokens**, **per-horizon loss normalization**, and **DDIM sampling** to balance predictive precision with calibrated price trajectories.
 - Developed a probabilistic evaluation suite optimizing PIT uniformity, interval coverage, and sharpness to **rigorously validate forecast reliability** before production delivery.
 - Owned the end-to-end pipeline from data ingestion to real-time serving and refined product strategy through direct feedback from **a16z** and **MetaProp**.
- Nov. 2024 – Present **Co-founder**, [PoliBOM](#), Seattle, WA
AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation
 - Specified an agentic workflow for BOM parsing and tariff simulation using Airflow, Elasticsearch embeddings, and FastAPI on Supabase and Redis.
 - Directed development of a **conversational AI interface** enabling decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
 - Executing customer discovery with ScoutyAI and Columbia economists, and secured early market validation by presenting the prototype to KPMG.
- Nov. 2023 – Present **Founder**, [Summit Geospatial](#), New York, NY
High-resolution elevation data and web delivery for engineering and hazard workflows.
 - Engineered a **statewide seamless Texas DEM mosaic**, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across **70+ state, federal, and local elevation datasets**.
 - Developed the web distribution platform end-to-end including **data pipeline, tiling, hosting, delivery UX** to support engineering and planning use cases.

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- Aug. 2021 – **Senior Data Scientist & Technical Lead**, [Texas Advanced Computing Center \(TACC\)](#), Austin, TX
- Aug. 2023
- Spun out Summit Geospatial alongside UT's VPs of IP to commercialize state-level terrain models from the \$40M [Texas Disaster Information System \(TDIS\)](#), delivering precision decision-support for civil engineering.
 - Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating [Real-Time Inundation Mapping](#) and TDIS.
 - Scaled climate and flood models on supercomputers, executing 800,000-node distributed jobs while managing \$1M+ compute budgets and federal partnerships with NOAA, NSF, USACE, GLO.
 - Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
 - Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
 - Contributed fundamental research supporting Paola Passalacqua's [2022 EGU Bagnold Medal](#), one of geomorphology's highest honors.
 - Served as Technical Reviewer for [Frontera Doctoral Fellows](#) (2023) and [Large-Scale Computing Pathways](#) (2022) applications, evaluating \$10M+ in computing resource allocations on one of the world's largest supercomputers.
- Feb. 2018 – **Data Scientist & Research Engineer**, [Texas Advanced Computing Center \(TACC\)](#), Austin, TX
- Jul. 2021
- Competed in [DARPA World Modelers](#) to improve disaster resiliency and food security modeling in East Africa.
 - Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
 - Received research recognition from AAI President Yolanda Gil during her [farewell address](#) for contributions to automated scientific modeling.

Research

- May 2025 – **Research assistant to Dir. Electrical Engineering Zoran Kostic**, [Columbia University](#), New York, NY
- Aug. 2025
- Math benchmarking and finetuning of Deepseek's GRPO algorithm
- Jan. 2025 – **Research assistant to Dir. Financial Engineering Ali Hirsra (Industry-sponsored research)**, [Columbia University](#), New York, NY
- Oct. 2025
- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
 - Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.
- July 2019 – **Summer research intern to Pres. AAI Yolanda Gil**, [University of Southern California](#), Los Angeles, CA
- Aug. 2019
- Integrated hydrological models into the [MINT Platform](#) during a year-long competition with MITRE to provide DARPA the "Airflow for science"

Teaching

- Jan. 2018 – **Co-instructor**, [The University of Texas at Austin](#), Austin, TX
- Dec. 2020
- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*.

Select Publications

- 2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TIS*, 2021. DOI: [10.1145/3453172](#)
- 2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: [10.1145/3308557.3308711](#)

Education

- expected 2026 **M.S., Urban Planning**, [Columbia University](#), New York City, NY
- 2017 **B.S., Pure Mathematics**, [University of Texas](#), Austin, TX

Key Projects

- oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.
- M&A predictor Built M&A prediction model (Oct.–Dec. 2025) with Prof. Eric Talley on 1.5M Compustat firm-quarters; long-only portfolio beat S&P 500 by 2%/yr (2004–2020), peaking at 6%/yr.

Activities

- Sailing Volunteer Atlantic SAR of the [Cheeki Rafiki](#) Bikepacking US West & East Coasts

Languages

- Spanish Native Proficiency French Professional Working Proficiency

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